Oxford Cambridge and RSA – style practice paper

GCSE (9–1) Computer Science

J277/01 Computer systems

Topic 1.1: Systems architecture

Candidate forename		**Candidate surname**	
Centre number		**Candidate number**	

Practice questions in the style of OCR GCSE Computer Science (J277). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 110 minutes (roughly 1 minute per mark)
Instructions	• Use black ink. HB pencil may be used for graphs and diagrams only. • Answer **all** the questions. • Write your answer to each question in the space provided. • Where the question asks you to write code or pseudocode, you may use any suitable high-level language or the OCR Reference Language; marks are for the logic, not exact syntax. • Do **not** use a calculator or any other electronic device unless told to.
Information	• The total mark for this paper is **100**. • The marks for each question are shown in brackets [ ]. • Quality of extended responses will be assessed in questions marked with an asterisk (*). • Where a question asks you to *state* or *give*, a short answer is enough; *describe* and *explain* need more detail.

Name: _____ Date: _____ Mark: _____ / 100

1 A computer uses a central processing unit (CPU) to run programs.

(a) State the purpose of the CPU. **[2]**

(b) Tick (✓) **one** box to show which type of computer architecture stores both instructions and data in the same memory. **[1]**

- ☐ Harvard
- ☐ Von Neumann
- ☐ Parallel
- ☐ Embedded

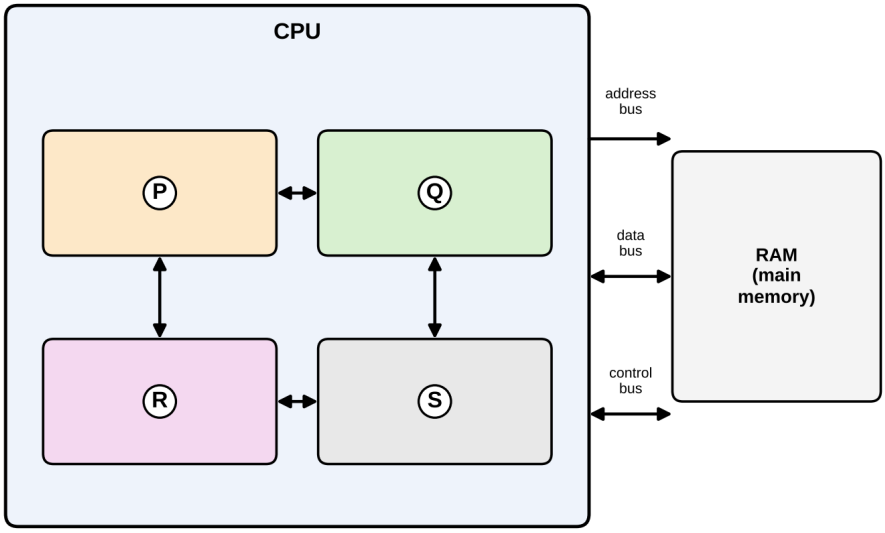
(c) The CPU is connected to main memory by buses. Name the bus that carries the memory address from the CPU. **[1]**

(d) Complete the sentences using words from the box. Each word can be used once. **[4]**

The _____ carries out arithmetic and logical operations.
The _____ sends signals to control the other parts of the CPU.
_____ is fast memory that stores frequently used instructions.
A _____ is a tiny storage location inside the CPU.

register	ALU	cache	control unit	RAM

2 The diagram shows the main parts of a computer system. The CPU contains four components, labelled **P**, **Q**, **R** and **S**.



(a) Component **P** sends control signals to the other components. Name component **P**. **[1]**

(b) Component **Q** performs calculations and compares values. Name component **Q**. **[1]**

(c) Component **S** is small and very fast memory that stores frequently used data and instructions. Name component **S**. **[1]**

(d) Component **R** contains the registers. Name **two** registers. **[2]**

(e) Explain why cache memory makes a computer run faster than if it only used RAM. **[3]**

(f) RAM is connected to the CPU by the address bus and the data bus. State what the **data bus** carries. **[1]**

3 Four registers are used in the fetch-decode-execute cycle.

(a) Draw **one** line from each register to its correct purpose. **[4]**

Register	**Purpose**
Program Counter (PC)	Stores the result of the latest calculation
Memory Address Register (MAR)	Holds the data that is being transferred to or from memory
Memory Data Register (MDR)	Holds the address of the next instruction to be fetched
Accumulator (ACC)	Holds the address of the memory location being accessed

(b) Describe what happens to the **program counter** during the fetch stage. **[2]**

4 The table shows the contents of part of the main memory of a simple computer. Each address holds one instruction. The program counter (PC) currently contains 2.

Address	**Contents**
1	LOAD 20
2	ADD 21
3	STORE 22
4	HALT
20	5
21	7
22	0

(a) State the address of the instruction that will be fetched next. **[1]**

(b) Complete the table to show the contents of the registers after the instruction at address 2 has been **fetched**. **[3]**

Register	Contents after the fetch
MAR	
MDR	
PC	

(c) The instruction at address 2 is **ADD 21**. It adds the value stored at address 21 to the value in the accumulator. Describe what happens during the **execute** stage. **[2]**

(d) Name the three stages of the fetch-decode-execute cycle in the correct order. **[3]**

5 Charlotte is comparing three computers. The table shows information about their CPUs.

Computer	Clock speed	Number of cores	Cache size
A	2.8 GHz	4	6 MB
B	3.6 GHz	2	4 MB
C	3.0 GHz	8	16 MB

(a) State which computer has the fastest clock speed. **[1]**

(b) Explain what is meant by a clock speed of 2.8 GHz. **[2]**

(c) Calculate how many clock cycles computer **B** completes in 5 seconds. Give your answer in billions. **[2]**

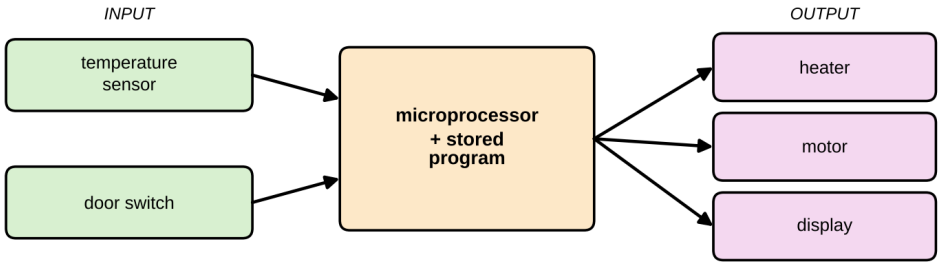
Answer _____ billion

(d) Computer **C** has 8 cores. Explain why this does not always make it twice as fast as a computer with 4 cores. **[2]**

(e) Charlotte plays games with complex 3D graphics. Recommend which computer she should buy. Justify your answer using the information in the table. **[3]**

(f) Give **one** reason why increasing the cache size can improve the performance of a CPU. **[1]**

6 A washing machine contains an embedded system. The diagram shows some of its inputs and outputs.



(a) What is meant by an **embedded system**? **[2]**

(b) Name **one** output of the washing machine system shown in the diagram. **[1]**

(c) Describe how the embedded system keeps the water at the correct temperature. **[3]**

(d) Give **two** differences between an embedded system and a general purpose computer such as a laptop. **[2]**

(e) Give **three** other examples of devices that contain an embedded system. **[3]**

7 Von Neumann architecture is used in most modern computers.

(a) State what is meant by the **stored program concept**. **[2]**

(b) Describe the role of the control unit in the fetch-decode-execute cycle. **[3]**

(c) State **two** things the ALU can do. **[2]**

(d) A program is stored on a hard disk drive. Explain why the program must be copied into RAM before it can be run. **[3]**

8 Some information about CPUs.

(a) A CPU has a clock speed of 3.2 GHz. Another CPU has a clock speed of 2.4 GHz. Calculate the percentage increase from 2.4 GHz to 3.2 GHz. Give your answer to the nearest whole number. **[3]**

Answer _____ %

(b) State **three** factors that affect the performance of a CPU. **[3]**

(c) Explain why a CPU with two cores is not always twice as fast as a CPU with one core, but may be faster when running several programs at once. **[3]**

9 Cache memory is located within the CPU.

(a) State **two** differences between cache and RAM. **[2]**

(b) Explain how a CPU uses cache when executing a loop that repeats the same 20 instructions 10 000 times. **[3]**

10 A smart thermostat is an embedded system that controls a home's heating.

(a) The thermostat uses a temperature sensor and a relay switch. State the type of device that is: (i) the sensor; (ii) the relay. **[2]**

(b) Explain why the program of an embedded system is usually stored in ROM or flash memory rather than on a hard disk. **[2]**

(c) Give **one** advantage and **one** disadvantage of using an embedded system rather than a general purpose computer to control the heating. **[2]**

11 Extended response questions.

(a) *Discuss how the clock speed, number of cores and cache size of a CPU affect the performance of a computer. In your answer, describe what each term means. **[8]**

(b) *Describe the fetch-decode-execute cycle, referring to the registers and components of the CPU that are used at each stage. **[8]**

END OF QUESTIONS

MARK SCHEME – Topic 1.1: Systems architecture

How to use this mark scheme: mark your own work in a different colour. A bullet (•) in the Answer column is a separate marking point worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed. **Accept** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. For programming questions, any valid solution that works is acceptable and syntax errors are not penalised if the intent is clear. The code in the answers uses the OCR Reference Language style.

Question	Answer	Marks	Guidance
1(a)	• to process data / carry out (execute) instructions ; • by repeatedly fetching, decoding and executing instructions (the FDE cycle) ;	2 AO1 1.1.1	Accept: it is the 'brain' of the computer only with extra detail.
1(b)	Von Neumann	1 AO1 1.1.1	
1(c)	address bus	1 AO1 1.1.1	
1(d)	• ALU ; • control unit ; • cache ; • register ;	4 AO1 1.1.1	
2(a)	control unit	1 AO2 1.1.1	
2(b)	arithmetic logic unit / ALU	1 AO2 1.1.1	
2(c)	cache	1 AO2 1.1.1	
2(d)	• program counter / PC ; • memory address register / MAR ; memory data register / MDR ; accumulator / ACC ;	2 AO1 1.1.1	Any two. Allow: MDR also called memory buffer register.
2(e)	• cache is faster to access than RAM ; • cache is located on / very close to the CPU ; • so the CPU spends less time waiting to fetch frequently used instructions / data ;	3 AO2 1.1.2	
2(f)	data / instructions between the CPU and memory ;	1 AO1 1.1.1	
3(a)	• PC: holds the address of the next instruction to be fetched ; • MAR: holds the address of the memory location being accessed ; • MDR: holds the data / instruction that has just been fetched from, or is about to be written to, memory ; • ACC: stores the result of calculations done by the ALU ;	4 AO1 1.1.1	Award one mark for each correct match. Accept correct purposes written next to the register.
3(b)	• the PC holds the address of the next instruction and this address is copied to the MAR ; • the PC is then incremented by 1 so it points to the next instruction ;	2 AO1 1.1.1	
4(a)	2	1 AO2 1.1.1	
4(b)	• MAR = 2 ; • MDR = ADD 21 ; • PC = 3 ;	3 AO2 1.1.1	Accept ADD 21 written as the instruction at address 2.

Question	Answer	Marks	Guidance
4(c)	• the control unit sends signals to fetch the value at address 21 ; • the ALU adds this value to the accumulator and the result is stored in the accumulator ;	2 AO2 1.1.1	
4(d)	• fetch ; • decode ; • execute ;	3 AO1 1.1.1	
5(a)	B	1 AO2 1.1.2	
5(b)	• the CPU carries out 2.8 billion cycles per second ; • each cycle can process one instruction / fetch-decode-execute step (approximately) ;	2 AO1 1.1.2	Accept 2 800 000 000 for 2.8 billion.
5(c)	• 3.6 × 5 ; • 18 billion (18 000 000 000) ;	2 AO2 1.1.2	
5(d)	• some programs are not written to use more than one core / cannot be split into independent tasks ; • cores may have to wait for each other / share the same memory / data ;	2 AO2 1.1.2	
5(e)	• recommend A, B or C (any, as long as justified) ; • reference to a feature, e.g. B has the highest clock speed / C has most cores ; • link to performance: more instructions per second / tasks in parallel / less waiting due to larger cache ;	3 AO3 1.1.2	
5(f)	more instructions / data can be stored close to the CPU so fewer slower accesses to RAM ;	1 AO1 1.1.2	
6(a)	• a computer system (processor / microprocessor) built into a larger device ; • designed to carry out a dedicated / specific task ;	2 AO1 1.1.3	
6(b)	heater / motor / display	1 AO2 1.1.3	
6(c)	• the temperature sensor sends the temperature (reading) to the processor ; • the processor compares it to the required value in its program ; • and switches the heater on / off accordingly ;	3 AO2 1.1.3	
6(d)	• embedded system: dedicated to one function / laptop: many functions ; • embedded program is stored permanently (ROM / firmware) / laptop can install new programs ; • embedded has no or limited user interaction ; embedded systems are small / low power ;	2 AO1 1.1.3	Any two differences.
6(e)	any three from: microwave oven ; central heating thermostat ; digital watch ; car engine management / ABS ; traffic lights ; alarm clock ; smart meter ; TV remote control ; printer ;	3 AO1 1.1.3	Do not accept laptop, desktop, tablet or smartphone (general purpose).
7(a)	• instructions and data are stored together in main memory / RAM ; • instructions are fetched one at a time from memory and executed ;	2 AO1 1.1.1	
7(b)	• fetches instructions from memory ; • decodes each instruction ; • sends control signals to the ALU / other components to execute it ;	3 AO1 1.1.1	

Question	Answer	Marks	Guidance
7(c)	• arithmetic operations (add, subtract, ...) ; • logical operations / comparisons (AND, OR, NOT, compare) ;	2 AO1 1.1.1	Also accept: binary shifts.
7(d)	• the CPU can only fetch instructions from main memory / RAM (not directly from secondary storage) ; • RAM is much faster to access than the hard disk ; • so the CPU does not have to wait / the program runs at an acceptable speed ;	3 AO2 1.1.1 / 1.2.1	
8(a)	• 3.2 − 2.4 = 0.8 ; • 0.8 ÷ 2.4 × 100 ; • 33 (%) ;	3 AO2 1.1.2	
8(b)	• clock speed ; • number of cores ; • cache size ;	3 AO1 1.1.2	
8(c)	• each core can process a separate instruction stream / program at the same time ; • so multitasking is quicker ; • but a single program may not be able to split into tasks that run in parallel (so no speed up) ;	3 AO3 1.1.2	
9(a)	• cache is faster (to access) than RAM ; • cache is smaller in capacity ; cache is more expensive per byte ; cache is inside / next to CPU while RAM is separate ;	2 AO1 1.1.2	Any two valid differences. Accept 'RAM is volatile' only if cache is also described as such.
9(b)	• the 20 instructions are copied into cache the first time ; • following repetitions fetch instructions from the (faster) cache instead of RAM ; • reducing time spent waiting for memory / improving performance ;	3 AO3 1.1.2	
10(a)	• sensor: input ; • relay: output ;	2 AO2 1.1.3	
10(b)	• the program is permanent and must not be lost when the power is off (non-volatile) ; • small / no moving parts / no need for an operating system ; (cheap, low power, robust)	2 AO2 1.1.3	
10(c)	• advantage: cheap / small / low power / reliable / fast response ; • disadvantage: cannot be easily upgraded / limited processing power / only does one job ;	2 AO3 1.1.3	

Question	Answer	Marks	Guidance
11(a)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • clock speed: number of cycles (instructions) per second measured in GHz; a higher clock speed means more instructions processed each second • very high clock speeds produce more heat and may need cooling / limits • cores: a core is a processing unit that can run its own fetch-decode-execute cycle; more cores allow several instructions / programs to be processed at once (multitasking / parallel processing) • benefit depends on the software: programs must be written to use multiple cores; otherwise little speed up • cache: very fast memory close to the CPU that stores frequently used instructions and data • bigger cache means more data available quickly so the CPU waits less for RAM; but it is expensive and has diminishing returns • performance is a combination: a CPU with a high clock speed but a small cache may spend time waiting; balanced judgement based on how the computer is used (gaming, office use)	8 AO1 / AO2 / AO3 1.1.2	

Question	Answer	Marks	Guidance
11(b)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • fetch: the address in the PC is copied to the MAR • the instruction at this address is fetched from memory (via the address / data buses) and placed in the MDR • the PC is incremented to point to the next instruction • decode: the control unit decodes the instruction held in the MDR to work out what operation is required and what data is needed • execute: the instruction is carried out; the control unit sends signals to the other components • ALU performs any calculation or comparison; the result is stored in the accumulator (or written back to memory via the MAR and MDR) • the cycle then repeats with the next instruction until the program stops • cache and clock may be mentioned for extra detail	8 AO1 1.1.1	

Where marks are lost in Topic 1.1

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Learn the four registers by what they hold:** PC = *address of the next instruction*; MAR = *address being accessed*; MDR = *data / instruction just fetched*; ACC = *result of ALU calculations*.
- ☐ **Fetch step:** PC → MAR, instruction → MDR, PC increments. Always say the PC is **incremented** — a favourite mark.
- ☐ **Performance questions:** do not just say 'faster'. Say *what* is faster: more cycles per second (clock), more instructions processed in parallel (cores), less waiting for RAM (cache).
- ☐ **Cores:** mention that software must be written to use them. 'Twice the cores = twice as fast' loses the mark.
- ☐ **Embedded systems:** dedicated function, built into a larger device. A laptop or smartphone is **not** embedded.
- ☐ **Clock speed:** 1 GHz = 1 billion cycles per second. Convert units before multiplying.
- ☐ **Von Neumann:** data and instructions are held in the **same** memory and fetched one at a time.
- ☐ **8-mark answers:** define each term, say the effect, then give a balanced conclusion. Aim for at least six distinct points.